Multiplicative Resonance Incidence at the No-Wrap Prime: Exact Character Spectrum, Parity Nullspace, and the Native Dispersion Gate

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Abstract

After the conductor-one sector is returned at the endpoint scale, the unresolved TPC-94 exceptional branch has algebraic conductor equal to the strong no-wrap prime q_X but insufficient archimedean variation on its affine interval. We isolate the exact finite-field geometry of this resonance.

For an odd prime q, put $G = \mathbb{F}_q^\times$ and

$$E_H = \{\pm 1, \dots, \pm H\} \subset G, \qquad 1 \le H \le (q-1)/2.$$

The resonance carrier is the multiplicative incidence operator

$$(K_H b)(\omega) = \sum_{r \in G} b(r) \mathbf{1}_{E_H}(r\omega).$$

We diagonalize it exactly by multiplicative characters. Its singular values are

$$\left| \widehat{\mathbf{1}_{E_H}}(\chi) \right| = \left| (1 + \overline{\chi(-1)}) \sum_{n \le H} \overline{\chi(n)} \right|.$$

Thus every odd character is an exact null mode. The principal singular value is $2H$, and

$$\sum_{\chi \ne \chi_0} |\widehat{\mathbf{1}_{E_H}}(\chi)|^2 = 2H(q-1-2H).$$

For nonprincipal even characters the Pólya–Vinogradov inequality gives $O(\sqrt{q} \log q)$.

Consequently, for arbitrary weights a, b on G, the incidence bilinear form has the exact principal/nonprincipal decomposition

$$\sum_{\omega, r} a(\omega) b(r) \mathbf{1}_{E_H}(r\omega) = \frac{2H}{q-1} \Big(\sum_{\omega} a(\omega) \Big) \Big(\sum_r b(r) \Big) + \mathcal{R}_H(a, b),$$

with

$$|\mathcal{R}_H(a, b)| \ll \sqrt{q} \log q \, \|a^\circ\|_2 \|b^\circ\|_2.$$

For a frequency-free literal branch $\beta = (\theta, c, \kappa)$ with resolved z-fiber length $1 \le N_\beta \le q_X$ and nonzero phase multiplier ω_β, resonance is exactly

$$r\omega_\beta \in E_{H_{q_X}(N_\beta)}, \qquad H_q(N) = \min\{(q-1)/2, \lfloor q/N \rfloor\}.$$

Branches with $N_\beta > q_X$ have no resonant nonzero frequency. This converts the resonant-sector census into a family of multiplicative spectral diagnostics and gives an explicit sufficient native-dispersion gate. A sharp delta-mass example proves that conductor q_X alone gives no saving. The paper proves exact L0 incidence theory and an L1 literal certificate interface, but no new signed Möbius estimate or L2 bound for the growing fixed-shift packet.

Keywords: affine Möbius correlation; archimedean resonance; multiplicative characters; incidence operator; no-wrap prime; exceptional sector; spectral gate.

Literal-frame convention. The physical interface always retains one prescribed nonzero h_0, both polarizations, every native key and exact-content child, the actual support and masks, and the original global normalization. The finite-field operator below controls the resonance incidence carrier; it is never substituted for the signed arithmetic sum without an explicit crosswalk.

1 Introduction

TPC-93 writes the literal low-frequency reconstruction term as a coherent sum of primitive fixed-determinant affine fibers [2]. TPC-94 computes the exact additive conductor: on literal support it is 1 or the no-wrap prime q_X [3]. TPC-98 returns the entire conductor-one sector at the critical amplitude scale $X^{o(1)}Q_X^2$ [4]. The next exceptional branch therefore has conductor q_X.

High conductor is not the same as visible oscillation. If a frequency-free resolved branch $\beta = (\theta, c, \kappa)$ contains N_β points in its z-fiber and its reduced phase is

$$e_{q_X}(-r\omega_\beta z), \qquad \omega_\beta \in \mathbb{F}_{q_X}^\times,$$

then the archimedean resonance condition is

$$N_\beta \left\| \frac{r\omega_\beta}{q_X} \right\| \le 1. \tag{1}$$

Even with $q_X \nmid r\omega_\beta$, this can hold throughout a short or arithmetically aligned interval.

The product $r\omega_\beta$ suggests using the multiplicative group $\mathbb{F}_{q_X}^\times$, not the additive Fourier transform that created the low window. The resonance window is symmetric about zero, and this symmetry has an exact spectral consequence: every odd multiplicative character is annihilated. The remaining even spectrum consists of classical incomplete character sums.

1.1 Main contributions

This paper proves:

1. the exact multiplicative convolution model for fixed resonance width H;
2. its complete character spectrum, principal mode, odd-character nullspace, and Hilbert–Schmidt identity;
3. exact principal/nonprincipal bilinear decompositions and L^2 bounds;
4. a positive literal resonance majorant with lossless frequency-free key coverage, grouped by native phase multiplier and exact width;
5. a sufficient spectral-dispersion condition that would return that carrier at $X^{o(1)}Q_X^2$; and
6. sharp concentration examples showing why conductor and cardinality alone cannot prove the condition.

The carrier is a positive or modulus-weighted majorant. Actual affine Möbius sums can retain r-dependent phases and signed weights. We therefore do not claim that character diagonalization alone evaluates the signed resonant contribution.

1.2 Why this is progress

Before this reduction, "resonance" was only a pointwise exception. The present result identifies its precise collective geometry. A future proof can now aim at one of two concrete targets: show that the native multiplier mass has a small principal component and controlled even-character spectrum, or exploit signed arithmetic before taking the incidence majorant. A proof-carrying polynomial obstruction to the displayed termwise bounds is a stop for that sufficient implementation, not a lower bound for the signed coefficient.

2 From archimedean resonance to a product window

Let q be an odd prime and identify each nonzero residue with its unique signed representative in $(-q/2, q/2)$. For $1 \le H \le (q-1)/2$, define

$$E_H = \{x \in \mathbb{F}_q^\times : |\widetilde{x}| \le H\} = \{\pm 1, \dots, \pm H\}. \tag{2}$$

Thus $|E_H| = 2H$.

For $1 \le N \le q$, put

$$H_q(N) = \min\left\{\frac{q-1}{2}, \left\lfloor \frac{q}{N} \right\rfloor\right\}. \tag{3}$$

Lemma 2.1 (Exact resonance window)**.** *Let $1 \le N \le q$ and $r, \omega \in \mathbb{F}_q^\times$. Then*

$$N \left\| \frac{\widetilde{r\omega}}{q} \right\| \le 1 \quad \Longleftrightarrow \quad r\omega \in E_{H_q(N)}.$$

Proof. For $N \ge 2$, the left side is equivalent to

$$|\widetilde{r\omega}| \le q/N.$$

The absolute residue is a positive integer, so this is equivalent to $|\widetilde{r\omega}| \le \lfloor q/N \rfloor = H_q(N)$. For $N = 1$, both sides hold for every nonzero residue because $H_q(1) = (q-1)/2$. □

Remark 2.2 (The range $N > q$)*.* If $N > q$, no nonzero residue can be resonant because $|\widetilde{r\omega}| \ge 1 > q/N$. Thus lemma 2.1 also recovers the conductor–length bound from TPC-94.

Definition 2.3 (Resonance incidence operator)*.* On $G = \mathbb{F}_q^\times$, set

$$(K_H b)(\omega) = \sum_{r \in G} b(r) \mathbf{1}_{E_H}(r\omega). \tag{4}$$

For $a, b : G \to \mathbb{C}$, define

$$\mathcal{I}_H(a, b) = \sum_{\omega \in G} a(\omega)(K_H b)(\omega). \tag{5}$$

Proposition 2.4 (Biregularity and total incidence)**.** *The bipartite graph*

$$r \sim \omega \quad \Longleftrightarrow \quad r\omega \in E_H$$

is $2H$-regular on both copies of G. In particular, for every $S \subseteq G$,

$$\sum_{\omega \in G} \#\{r \in S : r\omega \in E_H\} = 2H|S|.$$

Proof. For fixed r, multiplication by r is a bijection and exactly $2H$ values of ω send $r\omega$ into E_H. The same argument with the roles reversed proves both statements. □

3 Exact multiplicative character spectrum

Let $\widehat{G}$ be the group of multiplicative characters of G. For $f : G \to \mathbb{C}$, use

$$\widehat{f}(\chi) = \sum_{x \in G} f(x)\overline{\chi(x)}.$$

The inverse and Parseval formulas are

$$f(x) = \frac{1}{q-1} \sum_{\chi \in \widehat{G}} \widehat{f}(\chi)\chi(x), \qquad \sum_{\chi} |\widehat{f}(\chi)|^2 = (q-1) \sum_{x} |f(x)|^2.$$

Theorem 3.1 (Complete singular spectrum)**.** *The singular values of K_H are*

$$\left|\widehat{\mathbf{1}_{E_H}}(\chi)\right|, \qquad \chi \in \widehat{G}.$$

More precisely,

$$K_H\chi = \left(\sum_{x\in E_H} \chi(x)\right) \overline{\chi}.$$

Proof. For $\omega \in G$, substitute $x = r\omega$:

$$(K_H\chi)(\omega) = \sum_{x\in E_H} \chi(x\omega^{-1}) = \overline{\chi(\omega)} \sum_{x\in E_H} \chi(x).$$

Characters form an orthogonal basis, and conjugation permutes that basis. The claimed singular values follow. □

Theorem 3.2 (Parity nullspace)**.** *For every multiplicative character,*

$$\widehat{\mathbf{1}_{E_H}}(\chi) = (1 + \overline{\chi(-1)}) \sum_{n=1}^{H} \overline{\chi(n)}. \tag{6}$$

Hence every odd character, characterized by $\chi(-1) = -1$, is an exact zero singular mode of K_H.

Proof. Split E_H into $1, \ldots, H$ and their negatives. The negative half contributes $\overline{\chi(-1)} \sum_{n\le H} \overline{\chi(n)}$. □

Remark 3.3 (Not the sieve parity barrier)**.** The nullspace in theorem 3.2 is ordinary symmetry of a finite multiplicative convolution kernel. It is not a breach of the sieve parity barrier and gives no prime-pair lower bound.

Proposition 3.4 (Principal and Hilbert–Schmidt ledgers)**.** *For the principal character χ_0,*

$$\widehat{\mathbf{1}_{E_H}}(\chi_0) = 2H.$$

Moreover

$$\sum_{\chi\neq\chi_0} |\widehat{\mathbf{1}_{E_H}}(\chi)|^2 = 2H(q - 1 - 2H). \tag{7}$$

Proof. The first identity is $|E_H| = 2H$. Parseval gives

$$\sum_{\chi} |\widehat{\mathbf{1}_{E_H}}(\chi)|^2 = (q - 1)|E_H| = 2H(q - 1).$$

Subtract the principal square $4H^2$. □

Corollary 3.5 (Classical nonprincipal bound)**.** *Uniformly for $1 \le H \le (q - 1)/2$,*

$$\max_{\substack{\chi\neq\chi_0\\ \chi(-1)=1}} |\widehat{\mathbf{1}_{E_H}}(\chi)| \ll \sqrt{q}\log q.$$

When the indexing set is empty (in particular for $q = 3$), the maximum is understood to be 0.

Proof. Apply the Pólya–Vinogradov inequality to the incomplete character sum in (6) [1, Chapter 12]. □

4 Exact bilinear decompositions

For $f : G \to \mathbb{C}$, write

$$\overline{f} = \frac{1}{q-1} \sum_{x \in G} f(x), \qquad f^\circ = f - \overline{f}\,\mathbf{1}_G.$$

Theorem 4.1 (Principal/nonprincipal identity)**.** *For arbitrary $a, b : G \to \mathbb{C}$,*

$$\mathcal{I}_H(a,b) = \frac{2H}{q-1} \left(\sum_\omega a(\omega) \right) \left(\sum_r b(r) \right) + \mathcal{R}_H(a,b), \tag{8}$$

$$\mathcal{R}_H(a,b) = \frac{1}{q-1} \sum_{\chi \neq \chi_0} \widehat{\mathbf{1}_{E_H}}(\chi) \widehat{a}(\overline{\chi}) \widehat{b}(\overline{\chi}). \tag{9}$$

Only even nonprincipal characters contribute to (9).

Proof. Insert multiplicative Fourier inversion for $\mathbf{1}_{E_H}(r\omega)$ into (5). The principal term is the first term of (8). The parity assertion follows from theorem 3.2. □

Corollary 4.2 (Spectral norm bound)**.** *Let*

$$M_H = \max_{\chi \neq \chi_0} |\widehat{\mathbf{1}_{E_H}}(\chi)|.$$

Then

$$|\mathcal{R}_H(a,b)| \leq M_H \|a^\circ\|_2 \|b^\circ\|_2 \ll \sqrt{q} \log q \, \|a^\circ\|_2 \|b^\circ\|_2. \tag{10}$$

Proof. Use Cauchy–Schwarz in (9), followed by Parseval and corollary 3.5. Odd characters already vanish. □

Proposition 4.3 (Exact centered operator norm)**.**

$$\left\| K_H b - \frac{2H}{q-1} \left(\sum_r b(r) \right) \mathbf{1}_G \right\|_2 \leq M_H \|b^\circ\|_2.$$

The best constant is exactly M_H.

Proof. On the character basis, the centered operator deletes the principal mode and has the nonprincipal singular values from theorem 3.1. Their maximum is M_H, and an extremizing character gives equality. □

Remark 4.4 (Two distinct tasks). The principal term in (8) is a density term of size $2H/(q-1)$. The remainder is a multiplicative dispersion term. A proof must control both; a small nonprincipal spectrum cannot compensate for an unaffordable principal mass.

5 The literal native-multiplier interface

5.1 The frequency-free branch key

The resolved TPC-93 key $\xi = (\theta, c, \kappa, r)$ contains the low frequency r. It must not be inserted into a census and then paired with every frequency a second time. We therefore use the frequency-free TPC-94 branch

$$\beta = (\theta, c, \kappa).$$

Let $\mathfrak{B}_X^{\mathrm{nc}}$ be the finite set of soluble, nonempty literal branches β for which all inherited masks hold,

$$c \mid \Omega_\beta, \qquad q_X \nmid \Omega_\beta.$$

Throughout this section every sum over β is restricted to $\mathfrak{B}_X^{\mathrm{nc}}$. Here θ retains the polarization, opposite row, ℓ, j, σ, v, interval component, actual masks, projector labels, and all continuous separation data. It fixes the reflected ultra family, not one value of the opened divisor: the latter is $U_\theta(t)$ along the affine fiber.

For every soluble, nonempty branch put

$$b_\beta = c\kappa, \qquad N_\beta = \#I_\beta^*, \qquad \Omega_\beta = \ell_\theta v_\theta \sigma_\theta B_\beta.$$

Thus N_β is the cardinality of the resolved z-fiber I_β^*; it is not an orbit length in j. Let $A_{\beta,X}$ be the exact TPC-94 outer coefficient and let

$$\mathcal{A}_{\beta,X} = \sum_{z\in I_\beta^*} |\mathcal{W}_{\beta,X}(z)|\mu^2(\widetilde{D}_\beta(z))\mu^2(V_\beta(z)). \tag{11}$$

The exact branch sum satisfies

$$|S_{\beta,X}(r)| \le \mathcal{A}_{\beta,X} \quad \text{for every } r,$$

so the frequency-independent proof-carrying weight is

$$w_{\beta,X} = |A_{\beta,X}|\,\mathcal{A}_{\beta,X}. \tag{12}$$

This is the literal multiplier-mass return of TPC-94, not a new uniformity assumption [3].

On the nonconstant branch, literal provenance gives $c \mid \Omega_\beta$ and $q_X \nmid \Omega_\beta$. Define

$$\omega_\beta = \varepsilon_\theta \frac{\Omega_\beta}{c} \pmod{q_X} \in \mathbb{F}_{q_X}^\times. \tag{13}$$

The sign remains in the source key even though $E_H = -E_H$. Branches with $N_\beta > q_X$ have no resonant nonzero frequency by lemma 2.1; they contribute zero to the carrier below and are not discarded from the full signed formula.

For each exact width H, define

$$a_H(\omega) = \sum_{\substack{\beta:\ 1\le N_\beta \le q_X\\ H_{q_X}(N_\beta)=H\\ \omega_\beta=\omega}} w_{\beta,X}. \tag{14}$$

Equal multipliers are grouped only after every branch has received its own weight; the provenance bag remains recoverable.

Let

$$b_X(r) = |\mathfrak{m}_{K,X}(r)|\mathbf{1}_{\{0<d_{q_X}(r,0)\le R_X\}},$$

extended by zero to G. TPC-93 gives

$$\|b_X\|_1 \ll_K 1, \qquad \|b_X\|_2 \ll_K 1. \tag{15}$$

Definition 5.1 (Positive literal resonance carrier)*.* Set

$$\mathcal{C}^{\text{abs}}_{\text{res},X} = \sum_{\beta:\ 1\le N_\beta\le q_X} w_{\beta,X} \sum_{r\in G} b_X(r)\,\mathbf{1}_{E_{H_{q_X}(N_\beta)}}(r\omega_\beta). \tag{16}$$

Theorem 5.2 (Literal positive majorant and exact census identity)**.** *The absolute value of the signed nonconstant resonant contribution is at most $\mathcal{C}^{\text{abs}}_{\text{res},X}$, and*

$$\boxed{\mathcal{C}^{\text{abs}}_{\text{res},X} = \sum_{H=1}^{(q_X-1)/2} \mathcal{I}_H(a_H, b_X).} \tag{17}$$

For each H,

$$\mathcal{I}_H(a_H,b_X) = \frac{2H}{q_X-1}\|a_H\|_1\|b_X\|_1 + \mathcal{R}_H(a_H,b_X), \tag{18}$$

$$|\mathcal{R}_H(a_H,b_X)| \ll_K \sqrt{q_X}\log q_X\,\|a_H^\circ\|_2. \tag{19}$$

Proof. In the exact TPC-94 sector formula, the outer coefficient and branch mass obey

$$|A_{\beta,X}\mathfrak{m}_{K,X}(r)\zeta_{\beta,r}S_{\beta,X}(r)| \le w_{\beta,X}b_X(r).$$

The unit phase has disappeared only after taking absolute values. Lemma 2.1 supplies the indicator in (16); for $N_\beta > q_X$ it is identically zero. This proves the majorization without pairing one branch with any unintended frequency. Grouping the finite sum by the exact pair (H, ω_β) gives the identity (17). Now apply theorem 4.1 and corollary 4.2 and (15). ☐

Corollary 5.3 (A concrete sufficient gate). *The complete resonance carrier is $O(X^{o(1)}Q_X^2)$ if*

$$\sum_H \frac{H}{q_X}\|a_H\|_1 \ll X^{o(1)}Q_X^2, \tag{20}$$

$$\sum_H \|a_H^\circ\|_2 \ll \frac{X^{o(1)}Q_X^2}{\sqrt{q_X}\log q_X}. \tag{21}$$

Since $q_X \asymp Q_X$, the second target is

$$\sum_H \|a_H^\circ\|_2 \ll X^{o(1)}Q_X^{3/2}.$$

Proof. Sum (18) over H, use (15), and insert (20)–(21). ☐

Remark 5.4 (Sufficient, not necessary). The absolute carrier discards signed cancellation inside each affine sum and across native keys. Failure of (20) or (21) is not a lower bound for the carrier and does not disprove the original signed route. A proof-carrying polynomial obstruction in one of the actual termwise bounds only stops this sufficient principal-plus-absolute-remainder implementation; a sharper filter-sensitive estimate could retain cancellation between those pieces.

6 Sharp obstructions and stop tests

Example 6.1 (A full resonant low window at conductor q). *Take $\omega = 1$, $1 \le R \le H$, and let b be supported on $\{1, \dots, R\}$ with nonnegative values. Then*

$$r\omega = r \in E_H$$

for every active frequency, so

$$(K_H b)(1) = \|b\|_1.$$

The phase has algebraic conductor q for every nonzero r, yet the incidence gives no saving at all.

Proposition 6.2 (No conductor-only density bound). *There is no constant C, independent of q, H, such that*

$$(K_H b)(\omega) \le C\frac{H}{q}\|b\|_1$$

for all nonnegative b and all $\omega \in G$.

Proof. Use example 6.1 with $b = \mathbf{1}_{\{1\}}$. The left side is 1, whereas CH/q can tend to 0. ☐

Example 6.3 (Principal concentration). *If $a \equiv A$ with $A \ge 0$, and $b \ge 0$, then $a^\circ = 0$ and the nonprincipal remainder vanishes exactly, but*

$$\mathcal{I}_H(a, b) = \frac{2H}{q-1}\|a\|_1 \sum_r b(r).$$

Thus perfect multiplicative equidistribution does not remove the principal density ledger.

Example 6.4 (Odd-character null mode). *If $a = \chi$ is an odd multiplicative character and b is arbitrary, then the χ-component of the resonance carrier is annihilated exactly. This is the strongest possible spectral cancellation, but the literal positive envelope a_H need not lie in that subspace.*

6.1 Route decision

The incidence route continues only if a proof-carrying native census or theorem supplies both (20) and (21) for this termwise implementation. A proved polynomial obstruction in the principal bound calls for restoring cancellation before that split, or for a genuinely filter-sensitive positive theorem. An obstruction in the centered even-character bound calls for a second native averaging variable, the signed route, or the full-block route. These are stop tests for the displayed sufficient method, not lower bounds for the original signed coefficient. No amount of repeating "conductor q_X" changes them.

7 Exact finite regression

The script `experiments/tpc99_certificate.py` enumerates odd primes, all admissible widths H, and all multiplicative characters using a primitive root. It verifies:

1. exact resonance-window equivalence;
2. $2H$-biregularity and total edge counts;
3. the parity factor (6) and exact annihilation of every odd character;
4. the principal singular value and Hilbert–Schmidt identity (7);
5. direct incidence sums against their multiplicative Fourier reconstruction for deterministic complex test weights; and
6. the delta concentration obstruction; the direct Fourier reconstruction includes its principal contribution.

The spectral comparison uses high-precision floating arithmetic only to evaluate roots of unity. Graph, degree, cardinality, and resonance assertions are checked by exact integer arithmetic; character parity and complex annihilation are checked numerically in the primitive-root character enumeration. The certificate is a finite regression, not an asymptotic character-sum or Möbius theorem.

8 Route and exponent ledger

The TPC-98 constant-sector result and the present resonance operator give the literal exceptional split

$$\begin{gathered}\text{constant} \oplus \text{long nonconstant resonant} \oplus \text{long nonconstant generic} \\ \oplus \text{short nonconductor exceptions.}\end{gathered}$$

The first term is returned at $X^{o(1)}Q_X^2$. The second now has the two explicit gates (20)–(21). The third still requires a weighted two-form arithmetic estimate and coherent outer reassembly.

No exponent is double counted. In particular,

$$\lambda_D \le 2\eta_Z$$

remains the determinant/zero-mode compatibility, while

$$\Lambda_{\text{phys}} < \frac{1}{400}$$

remains the strict physical-loss ledger. Equality in the latter is a stop. The factor $\sqrt{q_X}\log q_X$ in (19) is part of the proposed resonance-dispersion route; it is not silently charged again if a future theorem proves (21) directly.

TPC-18 full-block reassembly and primitive determinant dispersion remain independent. An incidence-spectrum theorem on the low window proves neither.

9 Claim boundary

Unconditional results

- The resonance condition is exactly a multiplicative product window $r\omega \in E_H$.
- The resonance incidence graph is biregular and its complete singular spectrum is given by incomplete multiplicative character sums.
- All odd characters are exact null modes; the principal and nonprincipal Hilbert–Schmidt masses are explicit.
- The principal/nonprincipal bilinear identity and Pólya–Vinogradov L^2 bound are rigorous.
- The positive literal carrier and the two sufficient gates retain exact widths, multiplier magnitudes and support, both polarizations, and one global normalization. The phase of the reconstruction multiplier is deliberately lost only in the stated majorant.
- Conductor q_X alone cannot yield a pointwise or aggregate density saving without multiplier dispersion.

The finite-field results are L0. The native census and sufficient-gate crosswalk are L1 interfaces.

Not proved

- Neither (20) nor (21) is proved for the growing literal packet.
- The positive incidence carrier may be larger than the signed resonant contribution; no equivalence is asserted.
- No generic affine Möbius cancellation, determinant-energy lower bound, TPC-18 cover, fixed-shift Hardy–Littlewood asymptotic, parity breakthrough, prime-pair lower bound, or twin-prime conclusion is proved.
- Nothing is specialized to $h_0 = 2$.

Accordingly this paper sharpens the remaining door but does not claim an L2 arithmetic advance.

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